Lab 6

Exercises

1 a) $(1,3): a(1)^2 + b(1) + c = 3 \rightarrow a + b + c = 3$

$(2,5): a(2)^2 + b(2) + c = 5 \rightarrow 4a + 2b + c = 5$

$(3,3): a(3)^2 + b(3) + c = 3 \rightarrow 9a + 3b + c = 3$

$f[x_0, x_1] = \frac{5-3}{2-1} = 2$

$f[x_1, x_2] = \frac{3-5}{3-2} = -2$

$f[x_0, x_1, x_2] = \frac{-2-2}{3-1} = -\frac{4}{2} = -2$

$p_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$

$p_2(x) = 3 + 2(x-1) - 2(x-1) - 2(x-1)(x-2)$

$p_2(x) = -2x^2 + 8x - 3$

b) $p_2'(x) = 0$

$-4x + 8 = 0$

$4x = 8$

$x_{max} = 2$

c) $3(2^2 - 3^2) + 5(3^2 - 1^2) + 3(1^2 - 2^2)$

$= 3(4-9) + 5(9-1) + 3(1-4) = 16$

$2[3(2-3) + 5(3-1) + 3(1-2)] = 8$

$x_{max} = 16/8 = 2$ ✓

Computer lab

(a) x_i	$f(x_i)$	1st diff	2nd diff	3rd diff	4th diff
1	1				
		0			
2	1		0.5		
		1		1/3	
3	2		1.5		3/8
		4		11/6	
4	6		7		
		18			
5	24				

(c) Largest errors occur at the edges of the domain, specifically when $x < 1$ and $x > 5$ because the polynomial and cubic spline are forced to extrapolate outside the convex hull of the data, this is because of runge's phenomenon.

Initial AI Prompt Used

“Write a MATLAB script to interpolate data points for the gamma function and plot the results along with their absolute errors.

Here is the setup and the data:

X=[1,2,3,4,5]

Y=[1,1,2,6,24]

The evaluation domain for plotting is $x \in [0.5,5.5]$. Use 1000 points for a smooth curve”